VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-04 PHYSICS

Shailendra sir



Topics To Be Covered

1. TRANSMISSION OF POWER
2. POWER DISTRIBUTION TO A HOUSE
3. HOUSE WIRING (RING SYSTEM)
4. QUESTIONS


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Revision

Transformer → Mutual Induction

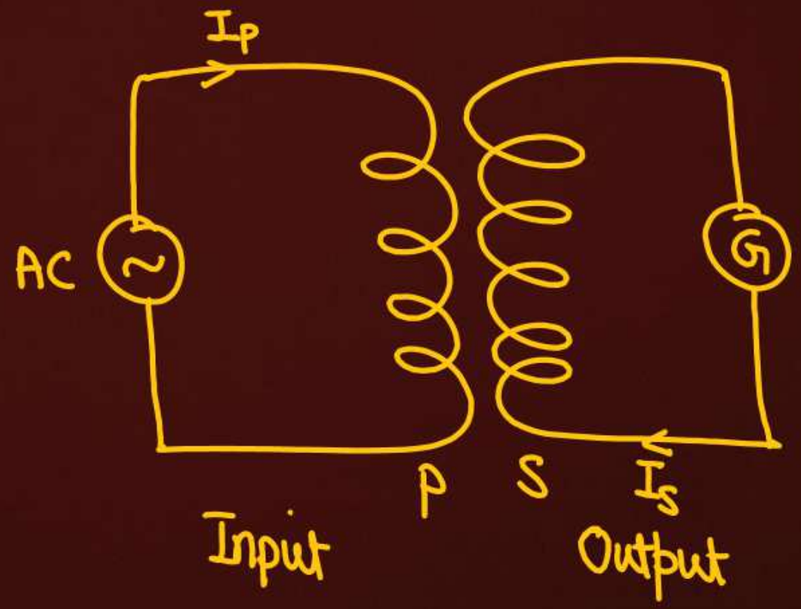
Power Generating Station → Home

V ↑ Insulation Problem

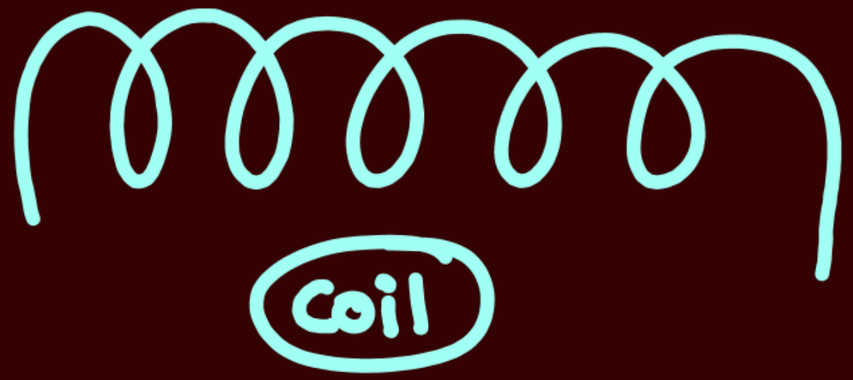
V ↓ Current increases. ⇓ Heat loss ↑

(AC.) Low Voltage → High Voltage
(Step Up Transformer) → $N_s > N_p$, $V_s > V_p$

High Voltage → Low Voltage
(Step Down Transformer) → $N_s < N_p$, $V_s < V_p$



$V_p \propto N_p$
$V_s \propto N_s$


coil



TRANSMISSION OF POWER FROM THE GENERATING STATION TO THE CONSUMER

- Electric power stations are generally located far from consumers.
- Power is generated at about 11 kV:
- Higher voltage → insulation difficulties
- Lower voltage → very high current
- AC power of 50 Hz is generated because AC voltage can be easily stepped up or stepped down using transformers.
- For long-distance transmission, voltage is stepped up above 11 kV.
- Higher transmission voltage → lower current → less heat loss in transmission wires.

$P_{\text{generating station}} = P_{\text{Home}} = \text{Constant}$

$VI = \text{Constant}$

$V \propto \frac{1}{I}$ for AC only



TRANSMISSION OF POWER FROM THE GENERATING STATION TO THE CONSUMER

Voltage Transformation

- **Generating Station:**
  11 kV → 132 kV using a step-up transformer
- ⬇ Long-distance transmission

- **Main Sub-station:**
  132 kV → 33 kV using a step-down transformer
  → Heavy industries & intermediate sub-station

- **Intermediate Sub-station**:
  33 kV → 11 kV using a step-down transformer
  → Light industries & city sub-station

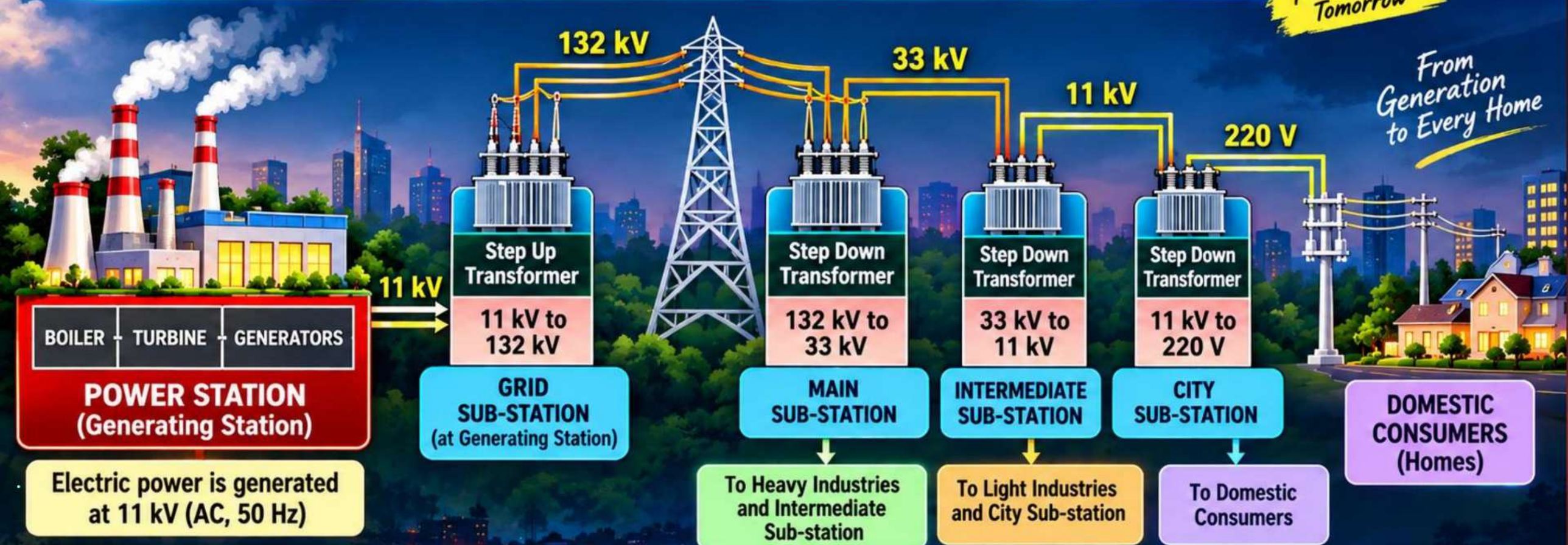
- **City Sub-station:**
  11 kV → 220 V using a step-down transformer
  → Domestic consumers

- Generating Station (11 kV)
- ↓
- Step Up (132 kV)
- ↓
- Main Sub-station (33 kV)
- ↓
- Intermediate Sub-station (11 kV)
- ↓
- City Sub-station (220 V)
- ↓
- Homes

TRANSMISSION OF POWER FROM THE GENERATING STATION TO THE CONSUMER

Electricity for a Brighter Tomorrow

From Generation to Every Home



Electric power is generated at 11 kV (AC, 50 Hz)

Higher voltage
Lower current

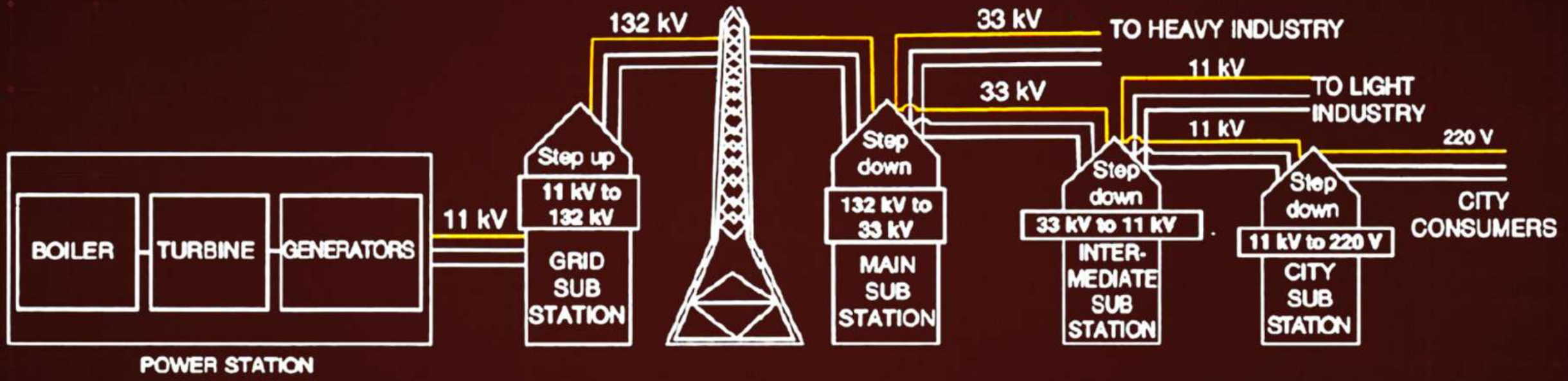
Less I^2R loss in transmission

Step-up & Step-down using Transformers

Safe and reliable power to homes

Powering People Powering Progress

TRANSMISSION OF POWER FROM THE GENERATING STATION TO THE CONSUMER



TRANSMISSION OF ELECTRICITY FROM POWER GENERATING STATION TO THE CONSUMERS

TRANSMISSION OF POWER FROM THE GENERATING STATION TO THE CONSUMER

Power generating station



at 11 kV

Grid sub station



step up to 132 kV

Main sub station

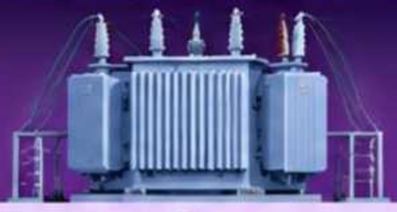


step down to 33 kV

To heavy industry



Intermediate sub station



step down to 11 kV

To light industry



City sub station



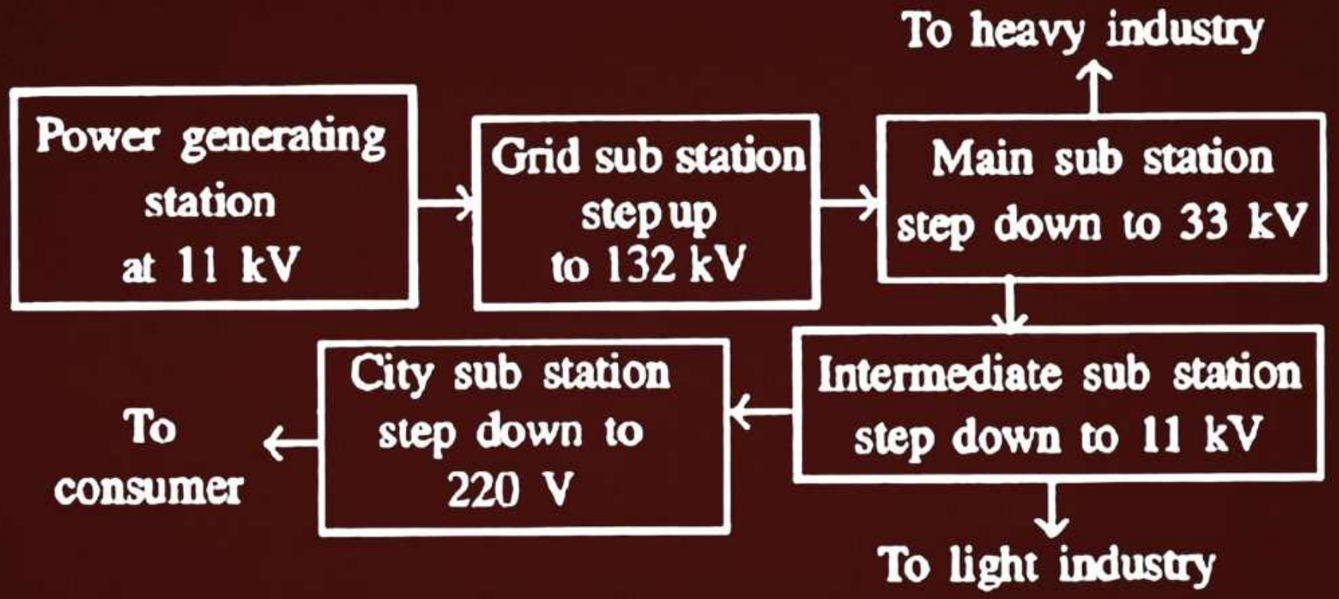
step down to 220 V

Consumer (Homes)



BLOCK DIAGRAM OF TRANSMISSION OF ELECTRICITY FROM POWER STATION TO THE CONSUMER

TRANSMISSION OF POWER FROM THE GENERATING STATION TO THE CONSUMER



BLOCK DIAGRAM OF TRANSMISSION OF ELECTRICITY FROM POWER STATION TO THE CONSUMER

QUESTIONS

The electric meter in a house records:

1. charge
2. current
3. energy
4. power

$E = P \; t$

$\downarrow \quad \downarrow$

$1\,\text{unit} = 1\,\text{kW} \; \text{h}$

$1\,\text{kWh} = 3.6 \times 10^6\,\text{J}$



POWER DISTRIBUTION TO A HOUSE

Domestic Electric Supply

- Electricity reaches homes from the city sub-station through overhead wires, cables on poles, or underground cables.

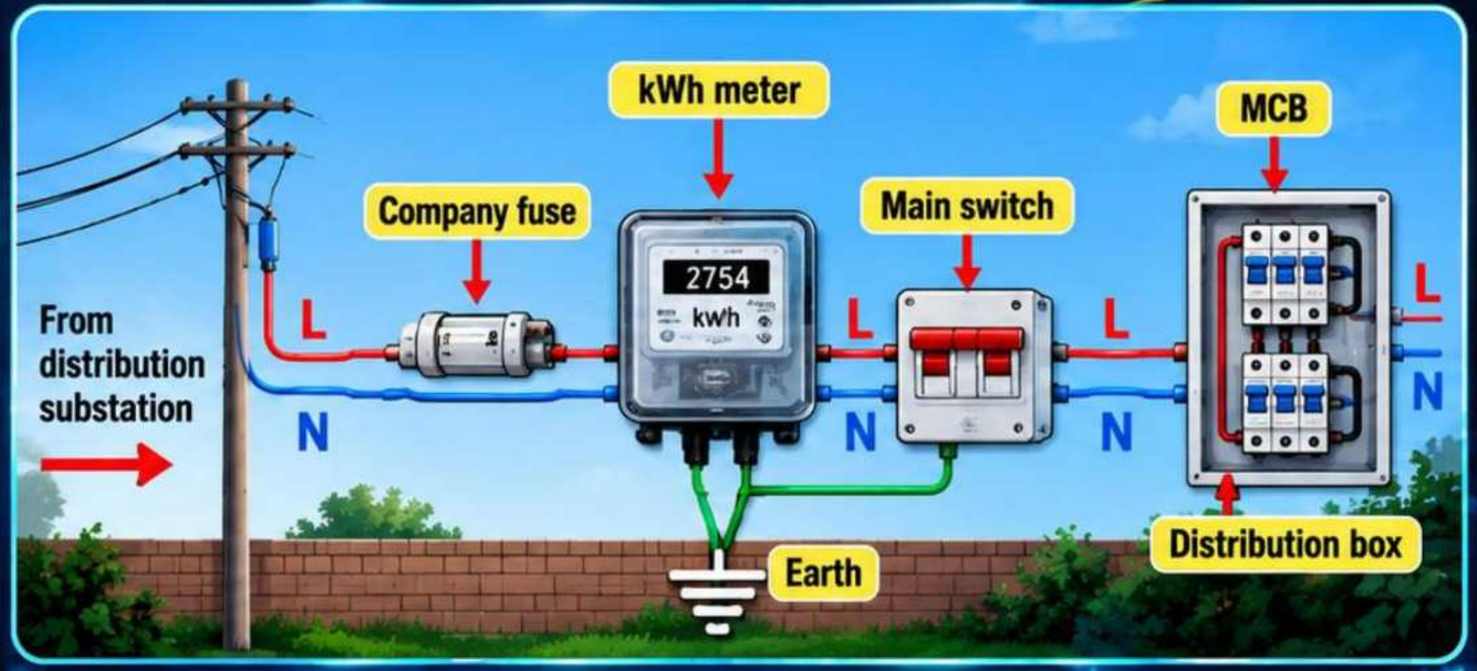
- A domestic supply cable has 3 wires:
- **Live/Phase Wire (L)** – carries current from the source to the house. 220V → Red
- **Neutral Wire (N)** – provides the return path for current. 0V → Black
- **Earth Wire (E)** – provides safety against electric shock. 0V → Green

- Neutral and Earth are connected at the local sub-station and maintained at approximately 0 V potential.

POWER DISTRIBUTION TO A HOUSE

Safe Power for a Brighter Tomorrow

- **A company fuse (pole fuse)** of high rating (≈ **50 A**) is connected in the live wire before the meter.
- It is handled only by **authorised electricity supply company personnel.** Its rating depends on the maximum load sanctioned for the connection.
- After the company fuse, the supply goes to the **kWh energy meter**.
- From the meter, it is connected to:
- **Main switch / ELCB** (earth leakage circuit breaker) → connected in both live and neutral wires.
- **Main fuse / MCB** (miniature circuit breaker) → connected only in the live wire.



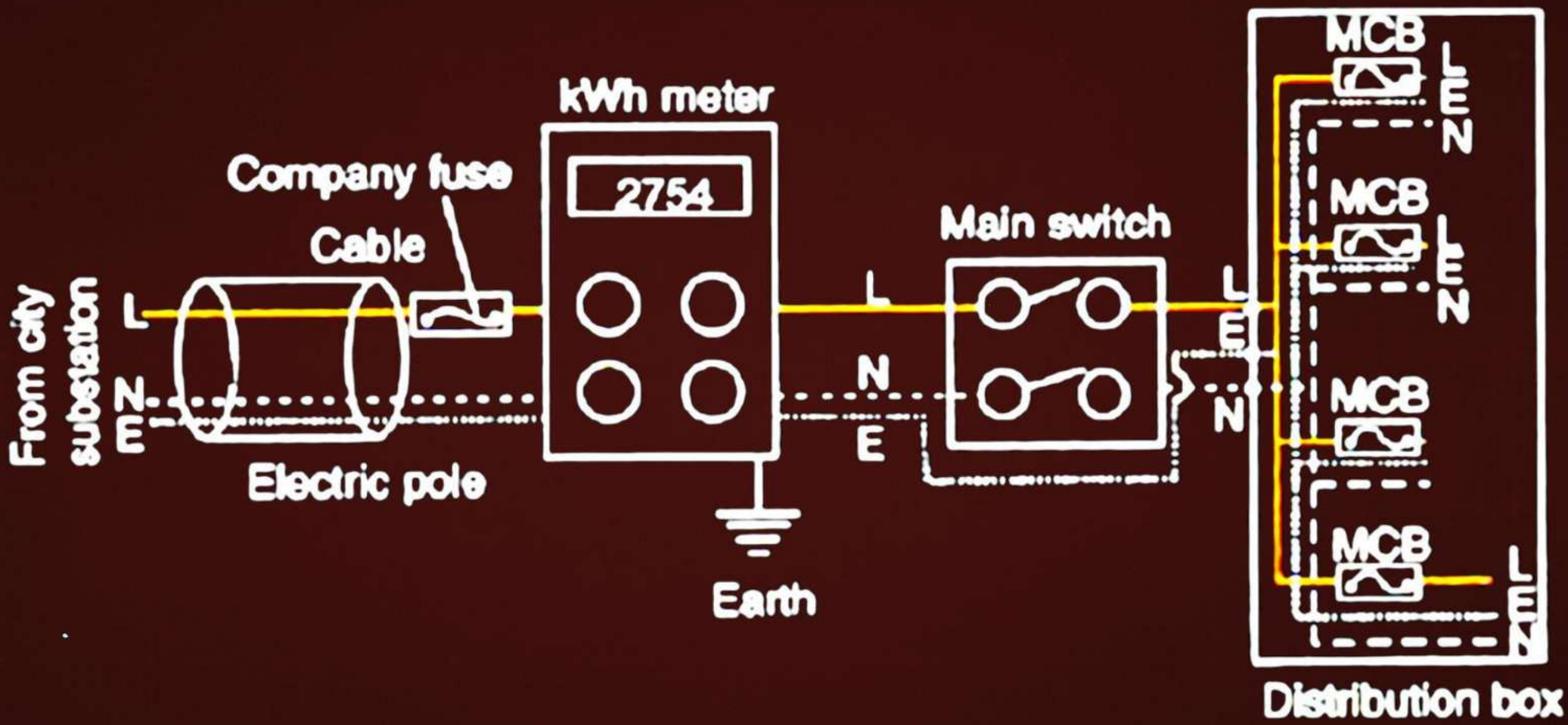
Pole Fuse → kWh Meter → Main Switch/ELCB → Main Fuse/MCB → House Circuits

SAFE AND RELIABLE POWER FOR EVERY HOME



Electricity Lights Lives

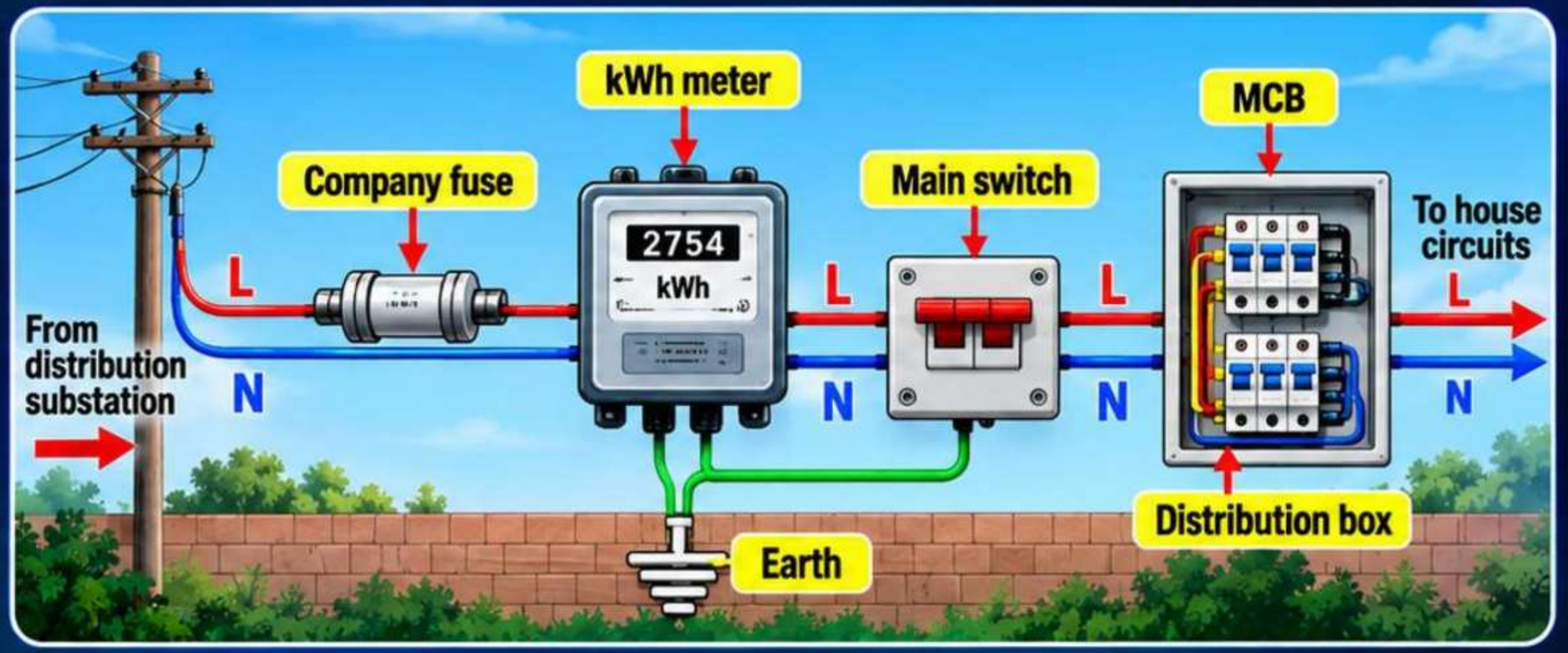
POWER DISTRIBUTION TO A HOUSE



POWER DISTRIBUTION TO A HOUSE

Electricity Reaches You Everyday

- Main switch is a **double-pole switch** that disconnects both live and neutral wires simultaneously.
- Its iron covering is **earthed for safety.**
- The **earth wire** from the meter is connected to an earth electrode in the house compound.
- From the distribution board, electrical supply is **distributed to different parts** of the house.
- Modern homes use a **consumer unit** after the kWh meter, containing:
  - Double-pole switch / ELCB (earth leakage circuit breaker)
  - MCB for each circuit



Key Point:

Main switch provides complete isolation of the house supply by disconnecting both **Live + Neutral** wires.



Powering Homes Powering Lives

SAFE DISTRIBUTION | RELIABLE SUPPLY | BRIGHTER TOMORROW



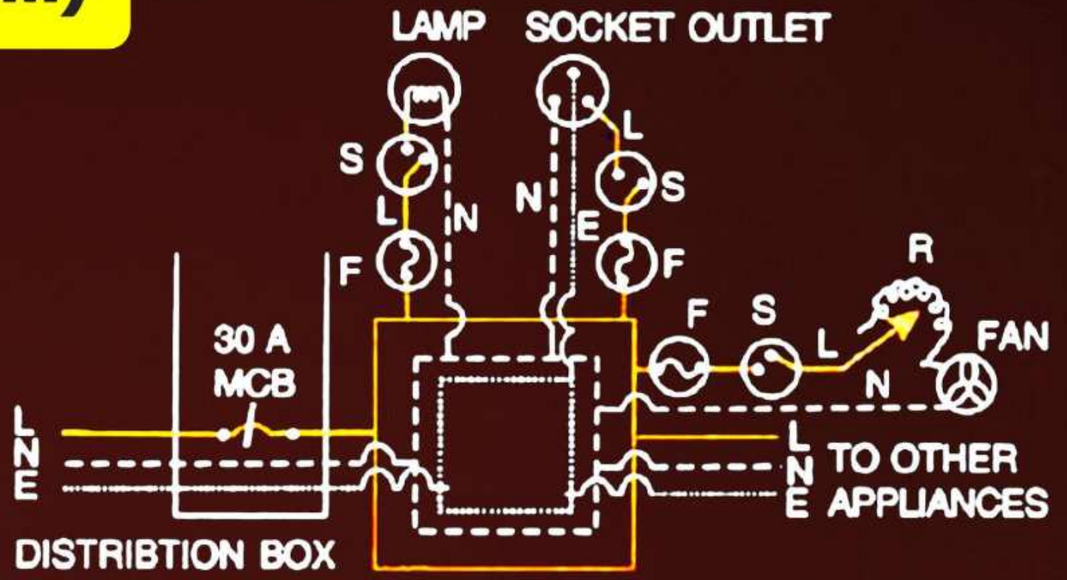
QUESTIONS

The main fuse is connected in:

1. live wire
2. neutral wire
3. both the live and earth wires
4. both the earth and neutral wires.

HOUSE WIRING (RING SYSTEM)

- House wiring is commonly done using the ring system. A separate ring is provided for each portion of the house.
- Wires from the distribution box run through all rooms and return to the distribution box, forming a closed ring.
- The distribution box generally has an MCB of about 30 A.
- Each appliance gets a separate connection from the live wire.
- The appliance is connected through a separate switch and fuse in the live wire.

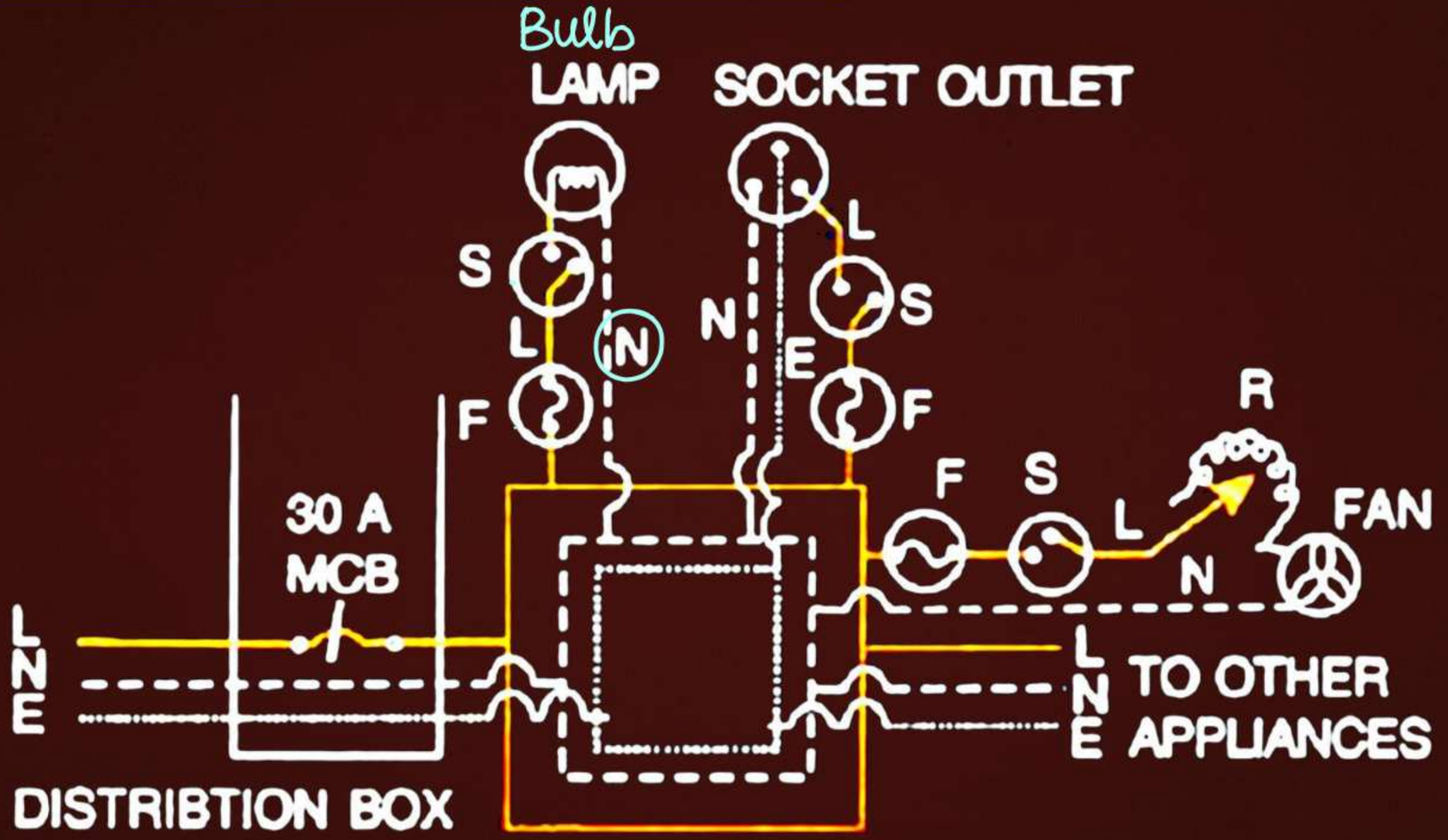


Ring system of house wiring showing circuit for lamp, socket and fan with fuse and switches in the live wire

- The other terminal of the appliance is connected directly to the neutral wire.
- The metal body/earth terminal of the appliance is connected to the earth wire for safety.
- Connecting wires must have adequate current-carrying capacity.

Live → Fuse + Switch → Appliance → Neutral Metal body → Earth wire

HOUSE WIRING (RING SYSTEM)



Ring system of house wiring showing circuit for lamp, socket and fan with fuse and switches in the live wire



ADVANTAGES OF THE RING SYSTEM

- In a ring system, current can reach an appliance through two separate paths.
- This makes each appliance effectively connected to mains by a thick wire.
- Hence, the main ring wire can be of lower current-carrying capacity → reduces wiring cost.

- Each appliance has its own fuse → fault in one appliance does not affect others.
- All plugs and sockets can be of the same size, but each socket must have a fuse suitable for its appliance.
- For a new appliance, no new line from the distribution box is needed → it can be directly connected to the ring circuit of the room.
- Care is taken that the total current drawn does not exceed the main fuse rating $(30A)$.



Advantages of connecting the appliances in parallel

- Each appliance gets connected to 220 V supply (= its voltage rating) for its normal working.
- Each appliance operates independently without being affected whether the other appliances are operated or not.



QUESTIONS

The electrical appliances in a house are connected in:

1. series
2. parallel
3. either in series or parallel
4. both in series and parallel.

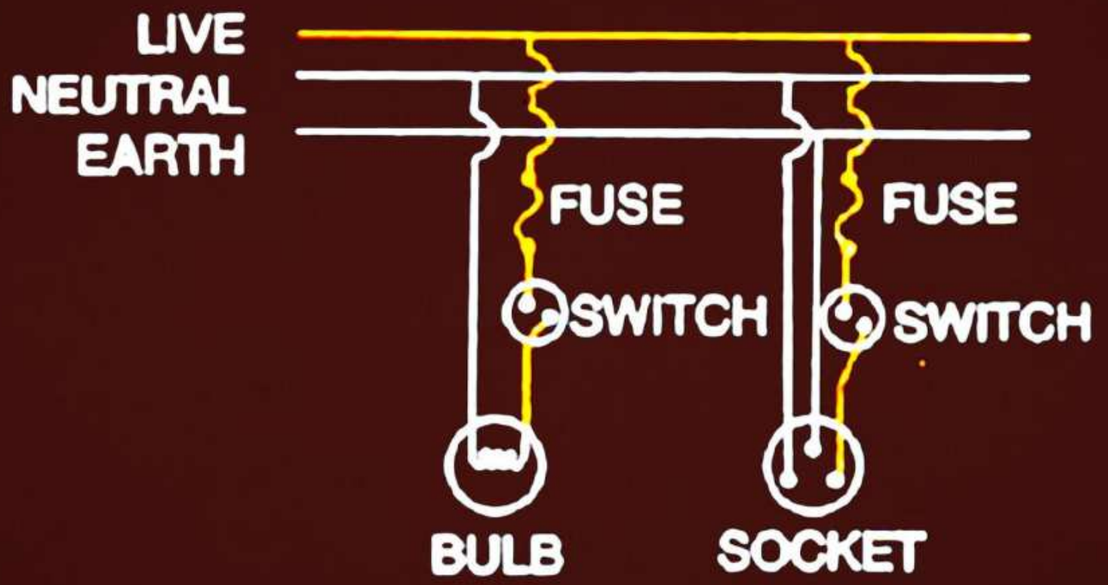


Disadvantages of connecting the appliances in series

- Voltage gets divided among appliances according to their resistances, so appliances may not receive their rated voltage.
- Adding another appliance increases the total resistance, which decreases the current and hence the power received by each appliance.
- Appliances cannot be operated independently in series.
- If one appliance is switched off or fails, the circuit breaks and all other appliances also stop working.

QUESTIONS

How will you connect a bulb and a socket in a room to the mains? Draw a labelled diagram in support of your answer. Use switch and fuse wherever necessary.



QUESTIONS

The diagram in Fig. shows a battery, a switch and two bulbs. (a) Complete the diagram to show the electric connections of the bulbs to the battery. (b) How have you joined the bulbs in part (a)? Give reason.

The two bulbs are connected in parallel.
Reason : (1) Both the bulbs glow at the same voltage. (2) If one bulb ceases to glow, the other bulb remains glowing.

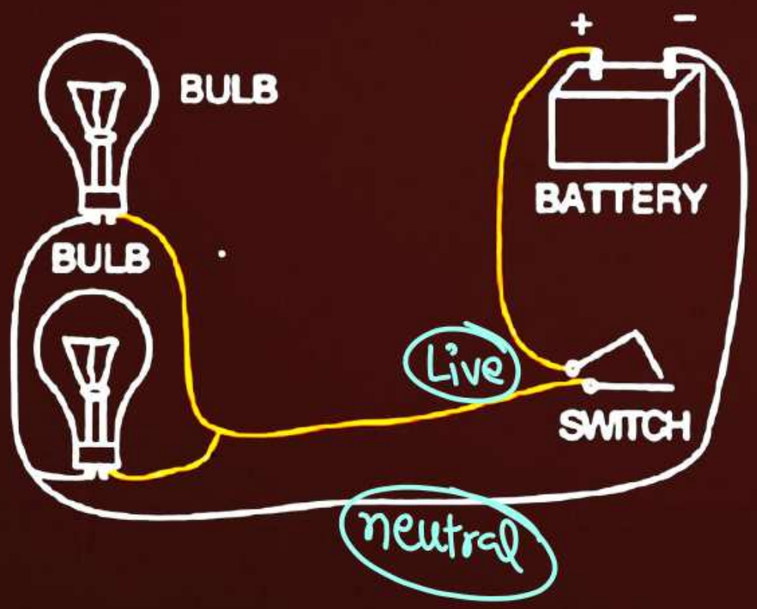




Fig. 9.6

QUESTIONS

Which of the following statement is incorrect regarding to step-down transformer?

$N_p > N_s$

$N_p > N_s$
$V_p > V_s$
$I_p < I_s$

1. number of turns in primary are more than the secondary coil
2. the primary coil is more heavily insulated as compared to secondary coil

$V_p > V_s$ $I_s > I_p$

3. the primary coil is thinner as compared to the secondary coil
4. the primary coil is thicker as compared to secondary coil



QUESTIONS

Kilowatt hour and kilowatt are:

energy Power

1. SI units of power and electric energy
2. commercial units of electric energy and power
3. commercial units of power and electric energy
4. SI units of electric energy and power



QUESTIONS

In a household electric circuit all appliances are connected in :

1. mixed circuit
2. parallel circuit
3. series circuit
4. any of these



FUSE

- The main fuse (or MCB) is placed in the live wire after the kWh meter and before the main circuit.
- Its job is to protect the circuit by breaking it when:
  - The circuit is overloaded (too many devices connected).
  - The circuit is short-circuited (direct contact between wires).
- Each electrical appliance also has its own fuse for safety.
- A fuse wire allows current to flow only up to a fixed limit (called current rating).
- If current goes above this limit:
  - The fuse wire heats up,
  - Reaches its melting point,
  - And then melts, breaking the circuit.

- **An electric fuse is a safety device which is used to limit the current in an electric circuit. The use of a fuse safeguards the circuit and the appliances connected in that circuit from being damaged.**



Principle

A fuse works on the heating effect of current. Due to flow of current I in it, heat produced per second in the fuse wire will be (In 1s.)

$$\mathbf{H = I^2R = I^2\rho l/\pi r^2}$$

If T is the temperature of surrounding of the fuse wire and ΔT is the rise in its temperature, loss of heat by radiation per second will be

$$\mathbf{H = 4\sigma T^3 \Delta T \times 2\pi r l}$$

where $\sigma(= 5.67 \times 10^{-8} \mathrm{Wm^{-2}K^{-4}})$ is a constant. In equilibrium,

$$\frac{I^2 \rho l}{\pi r^2} = 4\sigma T^3 \Delta T \times 2\pi r l$$

∴ The temperature of fuse wire above its surroundings is

$$\Delta T = \frac{\rho}{8\pi^2 \sigma T^3} \times \frac{I^2}{r^3} \quad \text{or} \quad \boxed{\mathbf{\Delta T \propto \frac{I^2}{r^3}}}$$

Thus, the rise in temperature of fuse wire is directly proportional to the square of current and inversely proportional to the cube of its radius. It does not depend on the length of the wire.



Principle

- A fuse is a short, thin wire having high resistance and a low melting point.
- It is designed to melt when excessive current flows, thereby protecting the circuit and appliances.

- Fuse wire is generally made of a lead–tin alloy because it has:
  - Low melting point ≈ 250°C
  - Higher resistivity than copper or aluminium.

- Higher current rating → thicker fuse wire.
  - 1 A fuse → radius 0.2 mm
  - 8 A fuse → radius 0.8 mm

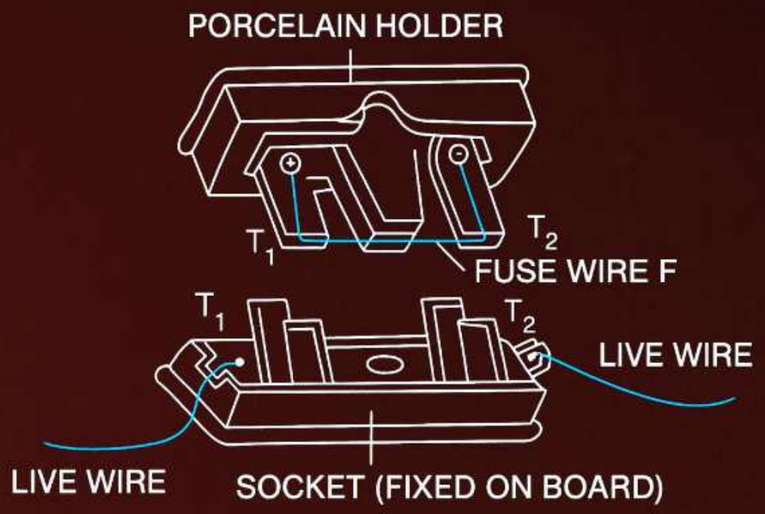
- **Copper and aluminium are unsuitable** because their melting points are very high (copper ≈ 1080°C) and ordinary wires are too thick to melt quickly during overcurrent.

Construction

Fig. shows one of the most common fuse arrangement in which the fuse wire F is stretched between the two metallic terminals T_1 and T_2 in a porcelain holder (since porcelain is an insulator of electricity).

- This holder fits into a porcelain socket having two metallic terminals to each of which the live wire of the circuit is connected. Thus, the fuse wire gets connected in the live wire.

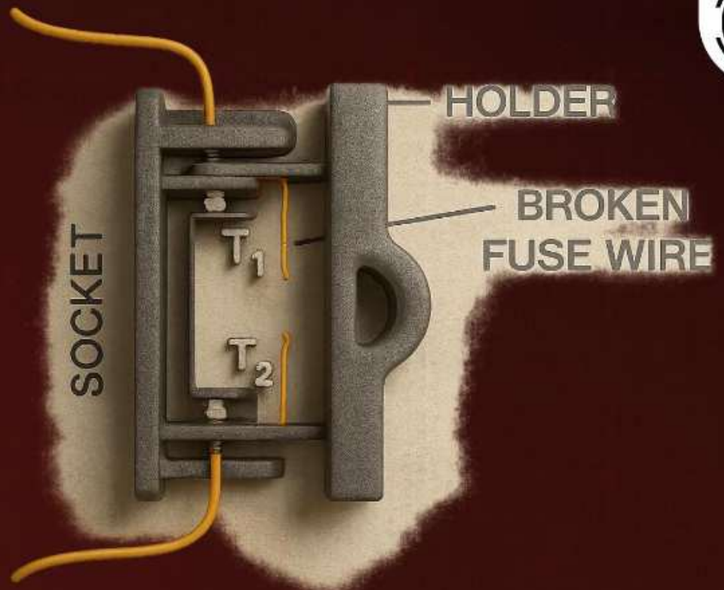


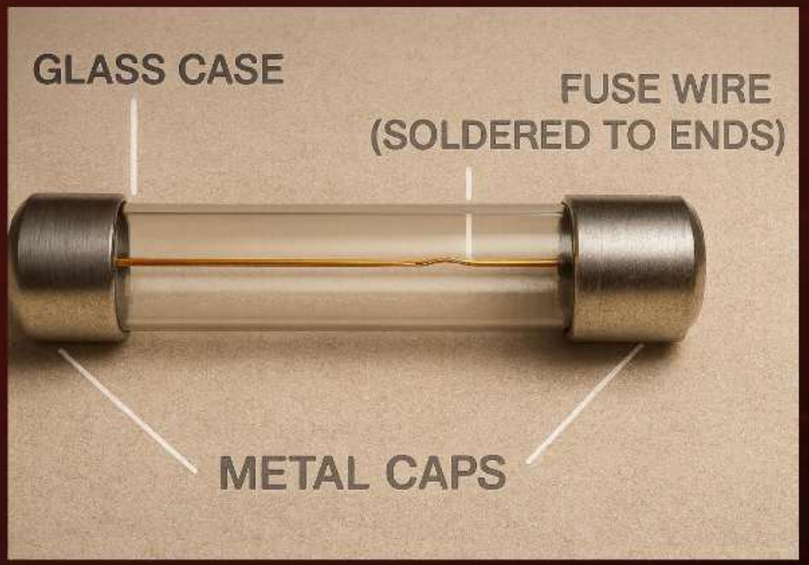




Working

- When current in the circuit exceeds the specified value (due to any reason such as high voltage, short circuiting, etc.) the temperature of the fuse wire rises to the extent that it melts.
- As a result, a gap is produced and the circuit breaks. Now current does not flow through the live wire and the appliance (or the circuit) is saved.
- After removing the fault in the circuit, a new fuse wire of the same current rating is connected in the holder to complete the circuit again.
- Now a days the costly appliances such as refrigerator, air conditioner, geyser, television, etc. are provided with a cartridge type fuse of proper current rating such as 1A, 2A, 5A, 10A and 13A.

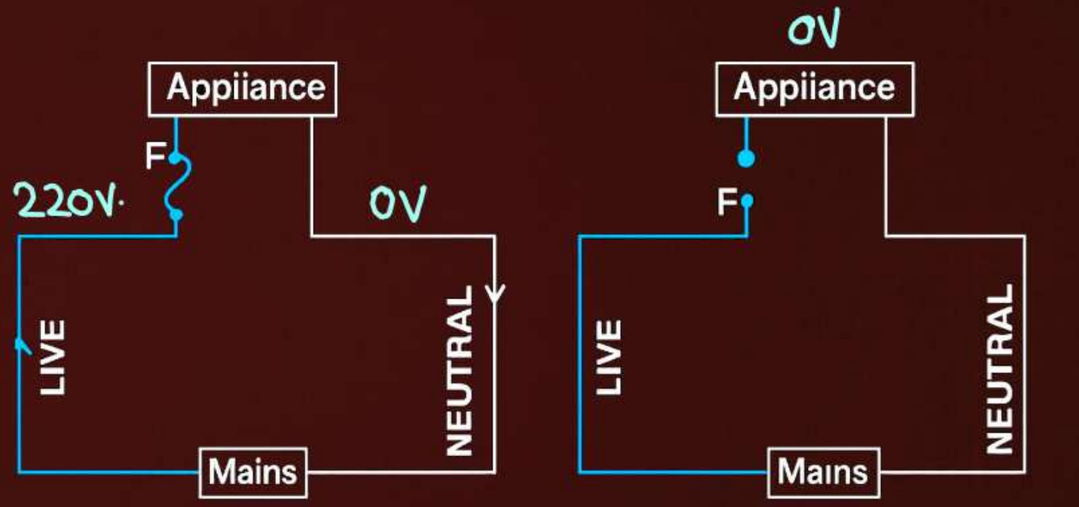




Reason for connecting the fuse in the live wire

The fuse is always connected in the live wire before the appliance, so that as the current in circuit exceeds the rating of fuse, it may melt and break the circuit first, before the current reaches the appliance. Thus, no current flows in the appliance.

In Fig, the fuse F is connected in the live wire joined to the appliance through the mains. The circuit is complete and current flows in the appliance. In Fig. (b), due to voltage fluctuation (or short circuiting), if excessive current flows, the fuse F blows off, the circuit becomes incomplete and no current flows in the appliance.

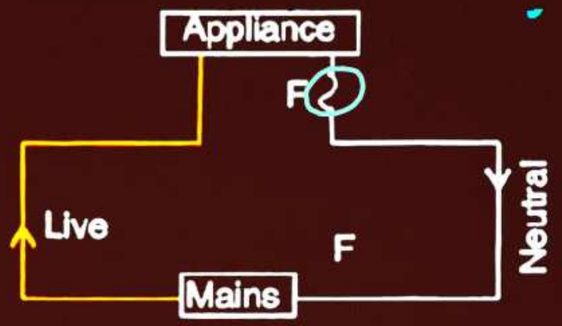


(a) Current flowing in the appliance through the fuse.

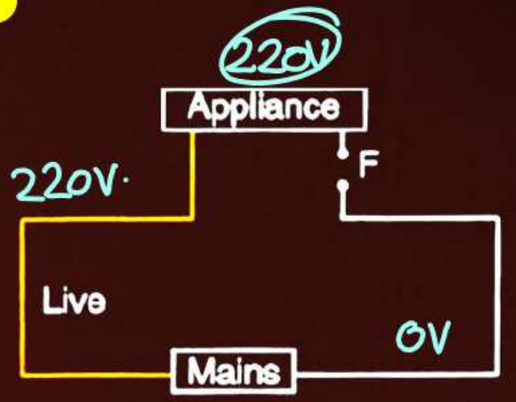
(b) No current flowing in appliance when the fuse blows off.

Reason for connecting the fuse in the live wire

- It is unsafe to connect the fuse in the neutral wire. Fig (a) shows the fuse F in the neutral wire. If due to some defect in the appliance an excessive current flows in the circuit, the fuse blows and current stops flowing in the circuit, but the appliance still remains connected to the high potential point of the supply through the live wire.



(a) Current flowing in the circuit



(b) Appliance connected to the high potential even when fuse blows off

- Now if a person touches the faulty appliance, he gets an electric shock because the person comes in direct contact of the mains through the live wire.
- Thus, it is highly unsafe to use fuse in the neutral wire.

HOMEWORK

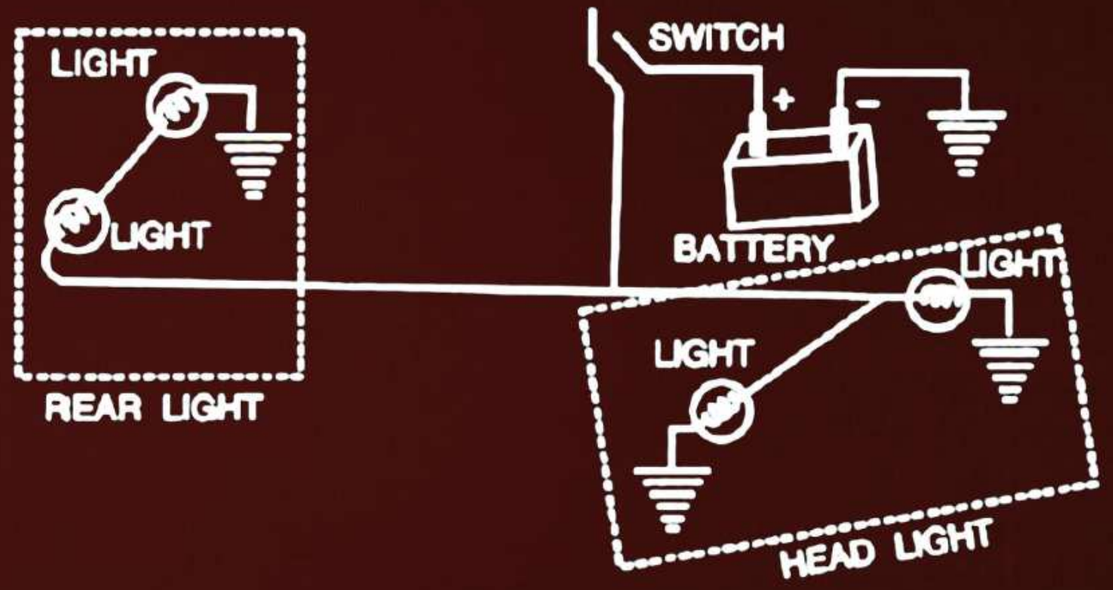
- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.

QUESTIONS

The diagram in Fig. shows the electrical system of a car to operate the two head lights and two rear lights by a switch. shows a connection to the body of the car.

(i) The diagram shows only one lead from the battery to each bulb, but a complete circuit must have two leads. How does the current get back to the battery?

(i) One lead of the battery and one terminal of the appliance (i.e., bulb) is connected to the body of the car (i.e., to the earth). The current gets back to the battery through the body of the car (i.e., earth). Thus the body of the car behaves like the other lead.



QUESTIONS

The diagram in Fig. shows the electrical system of a car to operate the two head lights and two rear lights by a switch. shows a connection to the body of the car.

(ii) The two rear lights as connected in diagram glow faintly. Why do they glow faintly? How should they be connected to glow brightly? Show by a separate diagram.

(ii) The two rear lights are connected in series so each light glows at half the voltage of the battery (if the resistance of filament of each bulb is same). Hence they glow faintly. They must be connected in parallel to glow properly as shown below in

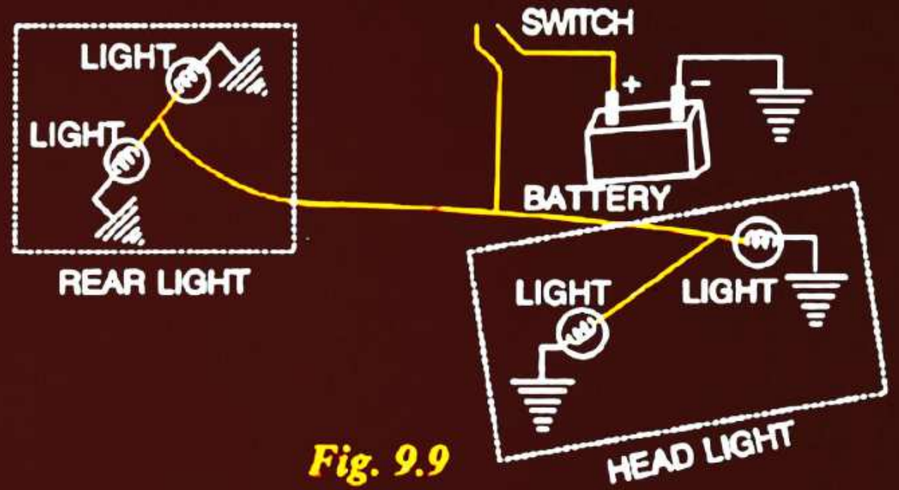
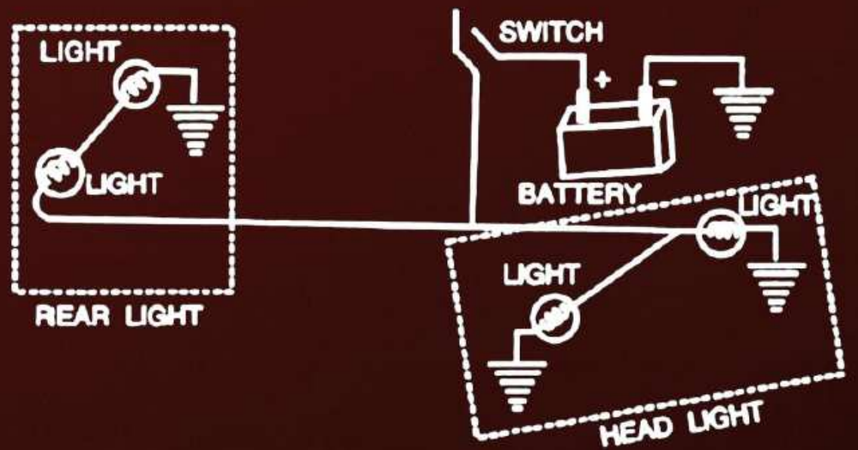


Fig. 9.9

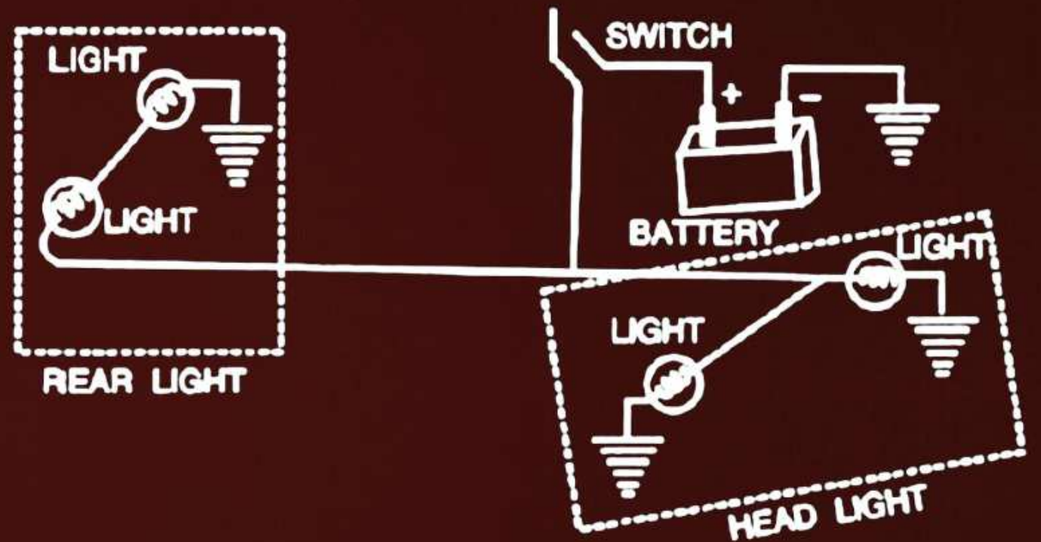


QUESTIONS

The diagram in Fig. shows the electrical system of a car to operate the two head lights and two rear lights by a switch. shows a connection to the body of the car.

(iii) If the lights are on, they become dim when the car is started. Give a reason.

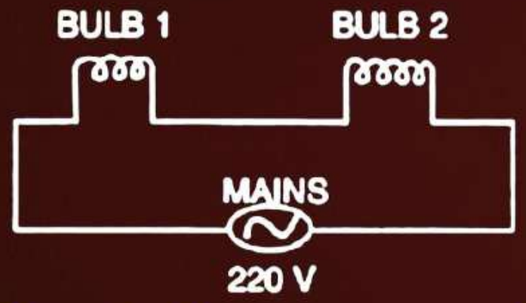
(iii) When the car is started, it draws heavy current from the battery. The terminal voltage of the battery falls due to voltage drop. As a result, the voltage across each bulb falls and they become dim.



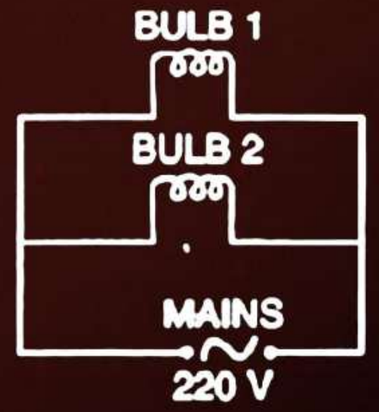
QUESTIONS

An electric bulb rated '$220\text{V}, 60\text{W}$' glows when connected with 220 V mains.
(i) Find the resistance of the filament of the bulb.

(ii) Another identical bulb is connected in series with the first one and the system is connected across the 220 V mains. Draw a diagram to show the arrangement and find: (a) the rate of consumption of energy in each bulb, and (ii) total power consumed.



(iii) If two bulbs are connected in parallel, draw a diagram of this arrangement. What will then be the total power consumed?



Thank You